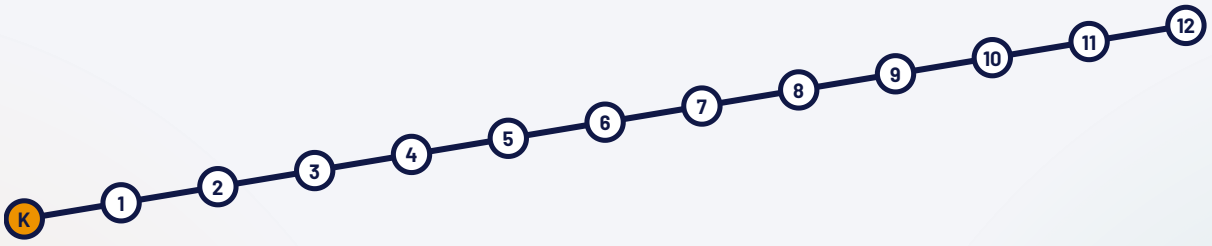
Awsaj Academy

K-12 VERTICAL ALIGNMENT · AN ANALYTICAL WALKTHROUGH

Our Curriculum Journey: Science

What our teachers' ratings tell us about how our standards are taught, K-12, and where we could go next.



From the ratings you entered as teachers (2,709 ratings, November 2026): what was reported as taught, not what students know.

Contents

1	**Science, K-12** Science from Kindergarten to Grade 12.	3
2	**About the data (at the back)** Where the data comes from, how to read the ratings, the words we use, the limits.	9

Each part: the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each. Items link to the Vertical Alignment site. About the data is at the back. 12 pages, A4.

Taught in depth Introduced / exposed Not taught

SCIENCE, KINDERGARTEN TO GRADE 12

Science, K-12

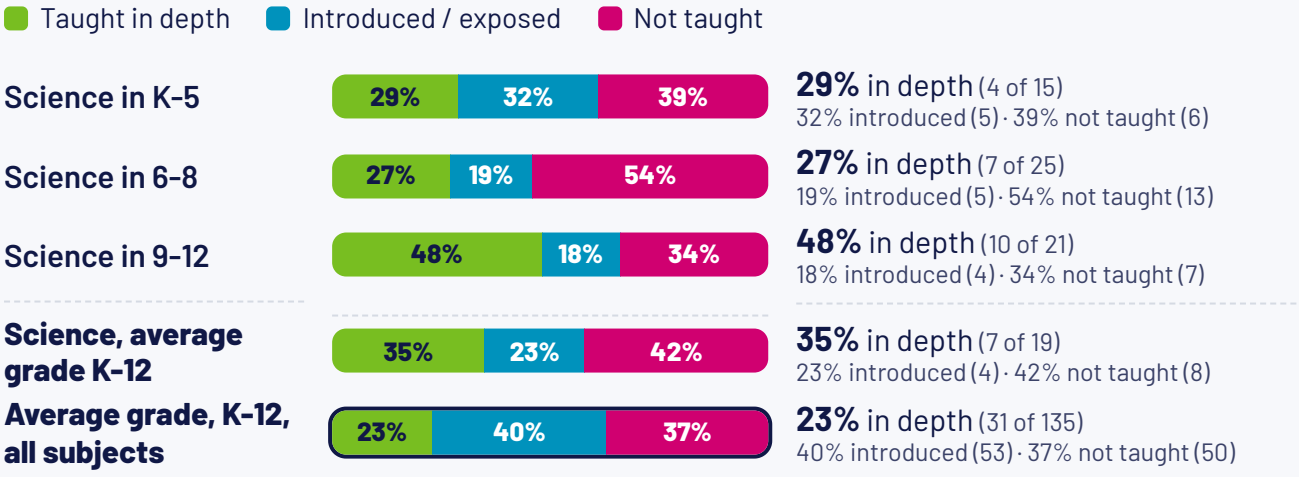
Across K-12, teachers reported that 35% of the 247 Science ratings (208 standards) are taught in depth, level with ELA as the highest in the school, and 58% are at least introduced. High school Physical Science (88% in depth) is one of the strongest strands in any subject.

THE PICTURE

At a glance: our curricular capacity

What you are seeing. An average grade in a year, by subject: green in depth, blue introduced, pink not taught, with the number of standards.

What it adds to our story. Science is often all or nothing: a topic is either taught deeply or not taught that year.



Curricular capacity: what an average grade reports taking to depth, introducing or not teaching in a year. From 247 grade-level ratings, Essential Elements apart. Not a target.

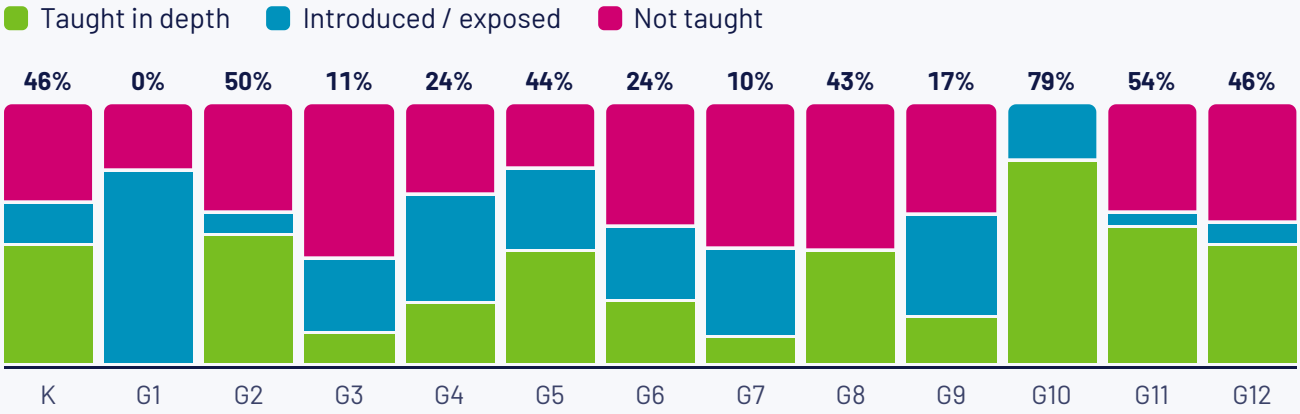
Open the coverage map on the site ↗

THE PICTURE

Grade by grade

What you are seeing. One column per grade. The top number is the share taught in depth.

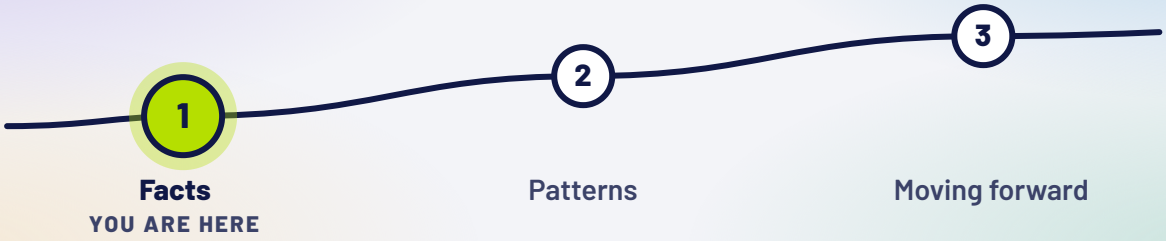
What it adds to our story. Depth changes from year to year, which may simply follow which units each grade runs.



	Elementary						Middle				High		
Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Earth and Space Science	40%	0%	100%	0%	0%	0%	67%	0%	0%	21%			
Engineering, Technology, and Applications of Science	33%	0%	0%	0%	0%	33%	0%	0%	100%	0%	0%	75%	0%
Life Science	100%	0%	100%	25%	100%	50%	0%	15%	0%			50%	
Physical Science	50%	0%	0%	0%	29%	83%	0%	0%	62%		100%		67%

How to read this: each square is the share of that strand's rated standards taught in depth in that grade. Darker green means more in depth. A blank square means the grade has no standards in that strand.

Open the grade dashboards ↗ Gaps and repetition ↗ Progressions ↗



SECTION 1 OF 3

Facts

What you are seeing. The five main facts. Each links to its data on the site.

What it adds to our story. When Science is taught, it is more often in depth than introduced. The open question is which year each topic lives in.

1. BRIGHT SPOTS **Physical Science builds to 88% in high school.** Physical Science is 31% in depth in K-5, 40% in 6-8 and 88% in 9-12. In Grade 10, all 15 Physical Science standards are in depth. Grade 10 dashboard ↗

2. BRIGHT SPOTS **Priority and power standards are deepest here.** 48% in depth, the highest of any subject: 37% in K-5, 50% in 6-8, 55% in 9-12. Grade 8 priorities are 85% in depth. Priority and power (Audit) ↗

3. BRIGHT SPOTS **K-5 progressions connect where units exist.** In K-5, 11 of 31 progressions are taught at every step, the most of any K-5 subject. 7 of 40 progressions break down on an official link, all in K-5. Progressions ↗

4. GAPS **Standards not taught in their own grade.** 24 confirmed, all in K-5: Physical Science 12, Earth and Space 8, Engineering 2, Life Science 2. In Grades 6 to 12, 49 more are rated not taught in the grade or course where they fit. Gaps and repetition ↗

5. GAPS **Hand-overs from K-5.** 17 standards taught in depth in Grades 6 to 10 (11 middle school, 6 high school) build on a K-5 standard rated not taught. All rest on suggested links, not official ones. Progressions ↗



Patterns

What you are seeing. What the facts show side by side. Possibilities to discuss, not conclusions.

What it adds to our story. Much of "not taught" in secondary Science is standards outside the discipline each grade teaches, and Science also carries a lot of literacy.

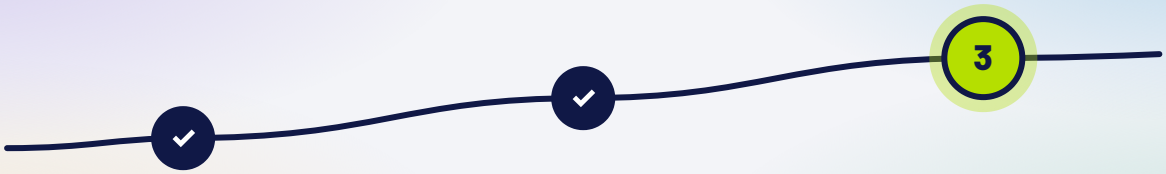
1. TRENDS ACROSS THE GRADES **Depth alternates from year to year.** High, low, high, low from Kindergarten to Grade 3, and again from Grade 6 to Grade 10. A unit-based plan could produce this pattern, which teams could confirm. Coverage map ↗

2. TRENDS ACROSS THE GRADES **Life Science has a long quiet stretch.** Strong in K-5 (100% in Grades 2 and 4), then 0% in Grades 6 and 8 and 15% in Grade 7, before returning in Grade 11 (50%). Coverage map ↗

3. CHOICES AND CAPACITY **Science is often all or nothing.** A standard is usually in depth or not taught, perhaps because Science is planned in whole units. It has the fewest "introduced only" standards (49, against 105 to 241 in the other subjects). Gaps and repetition ↗

4. CHOICES AND CAPACITY **Each secondary grade seems to own a discipline.** Grade 6 Earth and Space, Grade 8 Physical Science and Engineering, Grade 9 Earth and Space, Grade 10 Physical Science, Grade 11 Life Science, Grade 12 Physical Science. Standards outside a grade's discipline are rated not taught. Coverage map ↗

5. CHOICES AND CAPACITY **Engineering is taught in bursts.** It is 20% in depth K-12, with depth in Grade 8 (100%) and Grade 11 (75%). Shared across grade bands, it may be a strand no single grade feels it owns. Coverage map ↗



SECTION 3 OF 3

Moving forward

What you are seeing. Five questions for teams, each with its data and a first step we could take.

What it adds to our story. A shared K-12 map of Science units, so every strand has a home.

1 IMPROVING THE CURRICULUM **How might students keep meeting Life Science between Grade 5 and Grade 11?**

(Data: Life Science 0% in depth in Grades 6 and 8, 15% in Grade 7, 50% in Grade 11) see Pattern 2

First step: We might plan one Life Science unit for Grades 6 to 8.

Coverage map ↗

2 IMPROVING THE CURRICULUM **How will the 49 standards that fall outside each grade's main discipline be taught?**

(Data: 49 standards in Grades 6 to 12 not taught in the grade or course where they fit; no confirmed gaps in Grades 6 to 12) see Fact 1

First step: We could decide, course by course, which of the 49 standards to start teaching and which we choose not to use.

Gaps and repetition ↗

3 WORKING ACROSS GRADE LEVELS **What makes Grade 10 Physical Science work so well, and what could other courses borrow from it?**

(Data: Grade 10: all 15 Physical Science standards in depth, nothing not taught) see Fact 1

First step: We might ask Grade 10 Physical Science to share what makes its units work with the Grade 4, 5 and 8 teams.

Coverage map ↗

4

KNOWING THE BANDS ABOVE AND BELOW **Which K-5 Physical Science ideas do Grade 8 and Grade 10 most rely on, and how could the K-5 grades they belong to teach them?**

(Data: 12 K-5 Physical Science standards not taught in their own grade; 9 hand-over gaps run from them into Grades 8 and 10) see Pattern 4

First step: We could have each K-5 grade start teaching its own Physical Science standards that Grade 8 and Grade 10 build on.

Progressions ↗

5

HOW WE RATE AND PLAN **Do we have a K-12 map of which Science units run in which year, and does it match what the ratings show?**

(Data: Standards mostly either in depth or not taught; 49 introduced-only standards, the fewest of any subject) see Fact 4

First step: We could draw one K-12 map of which Science units run in which year, so every strand has a planned home.

Coverage map ↗

WHERE THE NUMBERS COME FROM AND HOW TO READ THEM

About the data

What our teachers reported about how our standards are taught, K-12. Eight parts: the whole school, the three grade bands and the four subjects. Each part opens with the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each.

ABOUT THE DATA

How to read this paper

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

What our teachers reported about how our standards are taught, K-12. Eight parts: the whole school, the three grade bands and the four subjects. Each part opens with the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each.

Where the numbers come from. The ratings sheet of 9 November 2026 (2,166 standards, 2,709 ratings), checked against the Vertical Alignment reports.

The three ratings.

Rating	What it means
Taught in depth	ALL classes in the grade consistently teach, reteach, review and assess the standard
Introduced / exposed	Anything in between (for example, 3 of 5 classes)
Not taught	No class in the grade teaches it

Keep in mind.

- The ratings report teaching across a whole grade, not what students know, and they judge no one.
- "In depth" is all or nothing. Teams may have read it more or less strictly.
- Much literacy is also taught in Science and Social Studies, and may not show in the ELA ratings.
- Shares are of grade-level ratings. Essential Elements are counted apart.

Curricular capacity. What an average grade reports taking to depth, introducing or not teaching in a year. Not a target: we cannot teach everything equally, so we choose the mix on purpose.

Categories. Each item carries a small label (for example Bright spots, Gaps, Handoffs), the same set in every part, so a team can follow one kind of observation across parts.

ABOUT THE DATA

Method, sources and limits

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

Every figure in this paper was computed from the ratings sheet and cross-checked against the site. Where the site's band and K-12 views differ (progressions), the paper says which it uses. Percentages are rounded half up, as on the site.

What was read

Source	How it was used	How fully it was read
Ratings sheet, "All Standards and Ratings K-12" (9 Nov 2026)	Every figure	All 2,166 rows
Organized ratings workbook	Cross-check of totals	All tabs
20 Findings decks (band and subject)	Cross-check, questions and next-step themes	All slides and speaker notes
K-12 Reports PDF (118 pages)	Cross-check	First 80 pages scanned by search; a few read directly
7 other Reports PDFs	Cross-check of headline figures	Scanned by search, not read page by page

How terms are used

- **Rated standards.** One rating is one standard in one grade. A standard shared by several grades is counted once for each grade that rated it.
- **At least introduced.** Rated in depth or introduced.
- **Not taught where it belongs (confirmed).** The grade the standard belongs to rated it not taught (for a standard shared across grades, every grade that shares it did). In Math and ELA each standard belongs to one grade, so this means that grade's team rated it not taught; it does not mean other grades should have taught it.

- **Not taught where it fits.** Each standard was placed in the grade or course where it fits best, and that grade or course rated it not taught. Other grades were not asked to rate it.
- **Introduced, not taken to depth.** Rated introduced by the grade (or grades) that rate it, and in depth by none. Most standards belong to one grade, so this counts what each grade introduces without taking to depth; it does not mean another grade should teach it again.
- **Progression finding.** An automatic check on a chain of linked standards: a step not taught, or a standard in depth while the one it builds on is not taught.
- **Taught at every step.** Every standard in the chain, across two or more grades, is at least introduced.
- **Progression.** A chain of linked standards across grades. Official links are the Coherence Map (Math), anchor standards (ELA) and the standards' own structure (Science, Social Studies). Other links are suggestions.
- **Curricular capacity.** The average number of standards one grade rated in depth, introduced and not taught in a year (all ratings divided by the number of grades), rounded.
- **Hand-over gap.** A standard taught in depth while a standard it builds on, placed in an earlier school, is rated not taught.
- **Priority and power standards.** Standards the school's own documents name: power standards in K-5, priority standards in 6-12.

Limits to keep in view

- Ratings are teachers' judgements of teaching across a grade. They are not observations and not student results.
- Grades may have read "in depth" differently. Uniform ratings across a whole grade are a reason to check how the scale was read; the paper points these out so teams can discuss them.
- Each team rated its own subject. Literacy taught in Science and Social Studies, and Math taught in Science, may not show in the ELA and Math ratings.
- Middle school Science and high school Math, Science and Social Studies standards were placed in the grade or course where each fits best. 24 K-5 Social Studies standards are also rated not taught in the grade where they fit.
- Band counts of progressions look only at steps inside the band, so they do not add up to the K-12 figure.

- Literacy audit figures come from the 2026 PK-5 literacy audit of weekly plans.
- Some strands have few standards in a grade, so a single rating can move a percentage a long way. Counts are given where this matters.
- Suggested links between standards (most ELA, Science and Social Studies hand-overs) have not been checked by teachers.

Status. This is a draft for the Vertical Alignment Committee. Each finding should be checked with the teams concerned before it is treated as settled. The planned review of weekly and unit plans (Part 2) will test many of these patterns against what is actually planned.